

LAST NAME: _____

Practice Quiz 03D — MATH 2055 — Fall 2026

Practice for Quiz 03

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let the universe of discourse be $D = \{0, 1\}$. Define predicates $P(x)$ and $Q(x)$ on D so that

$$\forall x \in D (P(x) \vee Q(x))$$

is true, but

$$(\forall x \in D P(x)) \vee (\forall x \in D Q(x))$$

is false. Give a two-row table listing $P(x)$ and $Q(x)$ for $x = 0, 1$, and use it to verify both claims.